

Auteur: Tala Mohanna
Studierichting: Informatica
Oefeningenreeks 1B nr. 17.21

$$(3 - i)^4$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(3 - i)^4 = 3^4 + 4 \cdot 3^3 \cdot (-i) + 6 \cdot 3^2 \cdot (-i)^2 + 4 \cdot 3 \cdot (-i)^3 + (-i)^4$$

$$= 81 - 108i + 6 \cdot 9 \cdot (-i)^2 + 12 \cdot (-i)^3 + 1$$

$$= 81 - 108i + 54 \cdot (-1) + 12i + 1$$

$$= (81 - 54 + 1) + (-108 + 12)i$$

$$= 28 - 96i$$

References